

Robotics in Healthcare:

Surgical and Rehabilitation Robots

A Research Report

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BS Artificial Intelligence
August 2026

1. Introduction

Robotics has moved from factory floors into hospitals and rehabilitation clinics over the last two decades. Surgeons now perform many minimally invasive procedures with robotic assistance, and stroke survivors regain arm and leg function with the help of robotic exoskeletons and therapy devices. It is easy to picture a robot working on a patient entirely on its own, but that is not what happens in most hospitals today. Almost every robotic device in clinical use requires direct human control or close supervision; the robot extends what a surgeon or therapist can do rather than replacing them.

This report focuses on robotic-assisted surgery and robotic rehabilitation: how these systems work, where they are used, their benefits and limitations, and the field's ethical challenges and future direction.

2. Robotics in Healthcare

Healthcare robotics covers a wide range of technologies, from robotic arms that hold cameras during surgery to companion robots used in elder care. Two categories dominate current clinical use: surgical robots, used mainly for greater precision and smaller incisions, and rehabilitation robots, used in physical therapy to help patients recover movement after conditions such as stroke, spinal cord injury, or orthopedic surgery.

Both categories share a feature that is easy to overlook: they are tools controlled or supervised by trained professionals, not independent decision-makers. It helps to distinguish three ideas: robotically assisted surgery, where a human surgeon controls the system directly; rehabilitation robots, which assist patients with repetitive, therapist-directed movement; and autonomous medical robots, which would plan and execute tasks with minimal human input. The third category remains a developing research area rather than routine practice (Lee et al., 2024).

3. Surgical Robots

How Surgical Robots Work

One of the most widely used surgical robotic platforms is the da Vinci Surgical System, developed by Intuitive Surgical. The surgeon sits at a console near the operating table and views a magnified, three-dimensional image of the surgical site. Hand movements are translated in real time into movements of small robotic arms holding instruments inside the patient's body, with tremor filtered out and hand motions scaled down into smaller, more precise instrument motions.

The robot does not decide where to cut or what tissue to remove. Every action originates from the surgeon's own hands; the system is a dexterous extension of the surgeon's control, not an independent operator (Buffi et al., 2025).

Major Applications

Robotic-assisted surgery is used across many specialties, including urology, gynecology, general surgery, cardiac surgery, and head and neck surgery, in procedures such as prostatectomy, hysterectomy, hernia repair, and cholecystectomy (gallbladder removal). Robotic platforms are especially valued in tight anatomical spaces, such as the pelvis, where wristed instruments can rotate in ways a human hand cannot. Newer competing systems, such as Medtronic's Hugo platform, have entered the market, giving hospitals more options beyond the established da Vinci system.

Advantages

Robotic-assisted surgery is generally performed through small incisions rather than large open cuts. This minimally invasive approach is associated with less blood loss, smaller scars, and often a shorter hospital stay, though outcomes vary by procedure and are not always superior to standard laparoscopic surgery. The three-dimensional view and tremor filtration help surgeons work with more consistency, and the improved instrument range of motion can make delicate steps, such as suturing deep inside the body, easier to complete.

Limitations

Robotic surgery also has real limitations that are sometimes underplayed. The equipment is expensive, often costing well over a million dollars, plus ongoing costs for instruments and maintenance. Operating times can be longer while a team is still building experience with the system, and the approach removes the surgeon's direct sense of touch, limiting the ability to feel tissue resistance.

Mechanical problems, while uncommon, do occur. A large pooled analysis covering more than three million procedures found a system malfunction rate of about one percent, with instrument-related faults most frequent; injuries linked to these faults were rare (Buffi et al., 2025). The FDA has also warned that safety and long-term cancer outcomes for robotic-assisted mastectomy and certain other cancer surgeries have not been established, a reminder that clearance for one procedure does not mean a device is safe for every use (U.S. Food and Drug Administration, 2022).

4. Rehabilitation Robots

Robotic Exoskeletons

Robotic exoskeletons are wearable devices with powered joints, usually at the hip and knee, that support or reproduce walking movements for people with reduced mobility. The ReWalk system is one well-documented example: it uses a backpack-mounted battery and computer, hip and knee motors, and a body-tilt sensor that detects when the user leans forward to trigger a step, with forearm crutches used for balance. ReWalk received its first U.S. clearance in 2014 through the FDA's de novo pathway for novel, moderate-risk devices, and later versions gained clearance for climbing stairs and curbs (U.S. Food and Drug Administration, n.d.). Exoskeletons such as ReWalk do not replace a wheelchair or cure the underlying injury; they work alongside a person's existing equipment and rehabilitation program, giving supervised opportunities to stand and walk (Christopher & Dana Reeve Foundation, 2025).

Robotic Therapy Systems

A second type of rehabilitation robot is used in clinics for repetitive movement therapy rather than walking outdoors. Electromechanical arm trainers guide a patient's arm through repeated reaching or grasping movements after a stroke, while robotic gait trainers support a patient's legs on a treadmill to practice a normal walking pattern. A therapist sets the resistance, range of motion, and repetitions, and monitors the patient throughout. These devices can deliver more repetitions per session than a therapist could usually provide manually, while recording data on strength and movement quality across visits.

Applications

Robotic rehabilitation is most established in stroke recovery, but it is also used after spinal cord injury, traumatic brain injury, and some orthopedic surgeries, ranging from large, fixed devices in hospital gyms to lighter wearable exoskeletons used in outpatient clinics or, increasingly, at home under remote supervision.

Benefits and Limitations

Evidence for robotic rehabilitation is more mixed than marketing materials sometimes suggest. A Cochrane systematic review of 45 trials involving over 1,600 stroke patients found that electromechanical and robot-assisted arm training produced a small but significant improvement in arm function and muscle strength compared with usual care, without increasing dropout or adverse events (Mehrholtz et al., 2018). A more recent meta-analysis of robot-assisted gait training after stroke likewise reported measurable benefits for walking function when combined with conventional physiotherapy (Lee & Kim, 2025).

At the same time, both reviews note that effects on broader daily activities are less consistent, and that trials vary widely in device type, training dose, and patient population, making sweeping claims about effectiveness hard to support. Cost and access are also genuine limitations: this equipment is expensive and remains concentrated in specialized centers, so patients in smaller clinics or rural areas may have limited access. As with surgical robots, rehabilitation robots support a clinician's plan of care rather than replacing one.

5. Benefits of Robotics in Healthcare

Across both surgery and rehabilitation, robotic systems share a few common benefits. They extend the precision and consistency of a trained professional, enabling fine or repetitive movements that would be tiring to sustain by hand alone. They can reduce physical strain on surgeons during long operations and on therapists guiding repetitive exercises. They also generate detailed data, such as instrument paths or joint movement ranges, which can support research, quality improvement, and more personalized therapy.

In surgery, these features often translate into smaller incisions and, for the right procedures, faster recovery, though results depend on the operation and the team's experience. In rehabilitation, robots make it feasible to deliver more intensive practice than limited therapy staff and time would otherwise allow, which matters given how repetition-dependent motor recovery tends to be. None of these

benefits are automatic; they depend on patient selection, proper training, and realistic expectations about what the technology can do.

6. Challenges and Ethical Considerations

Introducing robots into patient care raises challenges beyond the technical. Cost is an immediate concern: purchasing and maintaining these robots requires major investment, widening the gap between well-funded hospitals and smaller or rural facilities. Training is another concern, since staff need dedicated instruction and supervised practice before operating these systems safely; inadequate credentialing has been raised as a patient-safety issue in commentary following FDA communications about robotic surgery (U.S. Food and Drug Administration, 2022).

Questions of accountability also arise. If a robotic-assisted procedure leads to a complication, responsibility must be traced across the surgeon, the hospital, and the manufacturer, and this is not always straightforward. As surgical robots gain more automated features, including limited machine-learning capabilities in a small number of currently cleared devices, regulators and clinicians face new questions about how much decision-making can reasonably be delegated to software before meaningful human oversight is lost (Lee et al., 2024). Patient consent deserves attention too: patients should understand that "robotic surgery" almost always means a human-controlled system, not an independently operating robot, and that these systems increasingly collect procedural data that must be handled responsibly.

7. Future of Healthcare Robotics

Looking ahead, the clearest trend is a gradual, closely monitored increase in automated capability rather than a sudden jump to fully independent robots. A 2024 review of surgical robots cleared by the FDA between 2015 and 2023 found that most still operate at the most basic level of the researchers' proposed autonomy scale, meaning the surgeon remains in direct control of nearly every action, while only a small number of systems had reached a level involving conditional, task-specific autonomy under supervision (Lee et al., 2024). Fully autonomous medical robots remain an active research area rather than a clinical reality. In rehabilitation, the near-term direction points toward lighter, more affordable exoskeletons, telehealth-integrated remote supervision, and devices that adjust training difficulty automatically based on a patient's measured performance, within limits set by clinicians.

8. Conclusion

Robotic-assisted surgery and rehabilitation robotics have already changed how many operations and therapy sessions are carried out, but neither technology works independently of trained professionals. Surgical robots translate a surgeon's own hand movements with greater precision, while rehabilitation robots carry out a therapy plan designed and monitored by a clinician. Both offer genuine benefits, including steadier movement, higher repetition, and useful clinical data, but both also come with real costs, training requirements, and open questions about safety, access, and accountability. As the technology matures, and as automation is introduced carefully under regulatory review, healthcare

robotics is likely to keep expanding its role in patient care without replacing the judgment of the surgeons and therapists who direct it.

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